IIIT-Delhi Blog

Remote Hardware Labs in Covid Era: An Experiment at IIIT-Delhi

Image Source:https://www.reddit.com/r/FPGA/comments/i9pwgs/gives_me_a_headache_just_thinking_about_it/ Most of the universities around the world quickly moved to online teaching after the lockdown due to covid-19 pandemic. At IIIT-Delhi, the switch was not difficult since most of the courses involved software-based projects and assignments. However, IIIT-Delhi also offers hardware courses where students build complex embedded systems during weekly labs and facilitating the same in online mode was the challenging task. Over the years, students find hardware courses a bit difficult and time consuming which means we must make sure that students do not succumb to the excessive workload during these difficult times. Our team at Algorithms to Architectures Lab worked throughout the summer to identify various solutions for remote hardware labs and implemented the same during the Monsoon 2020 semester. Here we present some of our experiences, challenges and suggestions based on ECE270: Embedded Logic Design (ELD) Course with 92 students. The ELD course has three lectures and 2.5 hour lab per week and we try our best to keep the lecture and lab contents in sync with each other. For lectures, video recordings were shared before each session and students found that the overall ELD course was very well organised in its theoretical aspects. In the beginning, course instructor, Dr Sumit J Darak, informed that all the labs will be conducted remotely and there will not be any compromise on the learning outcome. To enable the same, IIITD ECE Lab took the responsibility of providing a 24/7 remote access to FPGA and SoC boards. For students with low-end laptops, dedicated desktops were also provided throughout the semester. The hardware servers were set up so that students can program the board remotely and verify the functionality on their laptop via logic analyzers. To help the learning, the classes and the labs both were kept in sync with each other from the beginning of the course. The teaching fellows (TFs) and teaching assistants (TAs) also worked exceptionally hard and pushed themselves to clarify our doubts. IIIT-Delhi's academic section also realized the workload at the instructor-end and helped us significantly by allocating additional TAs to support remote labs. The instructor, TAs, and TFs held regular office hours (there was at least one office hour per day) and often provided the necessary debugging help outside their office hours whenever needed. The lab handouts provided to students were too detailed and were more than enough to understand the concepts and complete the lab. For better understanding, we could always watch the labs' pre-recorded videos (around 60 mins per week), which were also very detailed. To reduce the workload, lab homework was removed and students were asked to complete the lab and appear for Viva every week (around 92 vivas were conducted in each week). Most of the students appreciated the idea of taking vivas as it ensured that the students take the labs seriously and only the honest and hard-working student gets the marks. However, we all know with everything right; there are always limitations and a room for improvement. To understand student's experiences and feedback, we, the student volunteers (details shared below) gathered suggestions from the rest of the ELD students. Most of the students agree with the fact that more number of boards could have been made available. For the students who use MacOS, the VIVADO software was not supported on their laptops, thus, pulling them back a little in the learning aspect. As there were a limited number of boards available for the students to work on, multiple students operated on the same board simultaneously, leading to incorrect results and creating a state of confusion. There could have been better communication among the students itself to avoid usage of the same board by multiple users at the same time to prevent chaos and confusion. As the complexity of the lab increased during the last few weeks, students got frustrated due to unavoidable glitches in remote access, other deadlines and increased workload. Thus, existing solutions for remote hardware access needs to be improved further. More importantly, students missed the opportunity to work on the hardware physically and explore integration of various sensors and displays. One interesting suggestion was to create live video demonstrations from ECE labs but this was not possible since TAs/TFs were working from home. Another suggestion is to plan additional lab sessions as and when campus opens. Indeed the ELD labs had

their ups and downs, but we appreciate that the ECE department did not choose

the easy path of dropping the hardware labs because the labs were vital and

integral to a student's learning experience. Written by IIITD ECE 2nd year students:Madhur Kumar, Samyak Gupta, Prashant Singh, Samaksh Gupta,Saksham Gupta, Tushar Agarwal, Divin Dominic, Mohd. Siraj Ansari, Rishi SinghalSupervised By:Dr. Sumit J Darak (Associate Professor and Chair: UG Affairs, ECE, IIIT Delhi)

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